

UNIVERSITY OF SANTO TOMAS
Faculty of Engineering
ELECTRONICS ENGINEERING DEPARTMENT
Second Term, AY 2025 – 2026



ECE21120L: Electronics Systems and Design, Laboratory

Laboratory Experiment No. 9
PROGRAMMABLE LOGIC CONTROLLER

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I. OBJECTIVE

1. To familiarize and learn how to use the OpenPLC editor software.
2. To learn the basics of ladder diagram and simulate logic gates using OpenPLC editor software.
3. To use OpenPLC and construct a ladder diagram based on a specific logical requirement.

II. MATERIALS AND EQUIPMENT

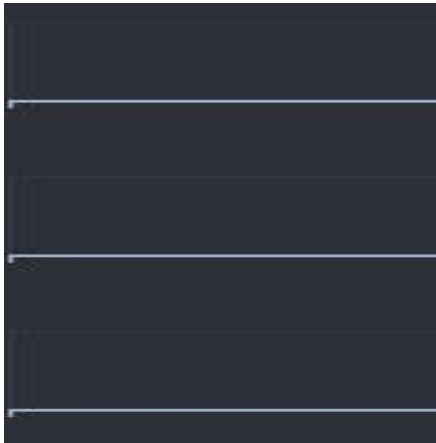
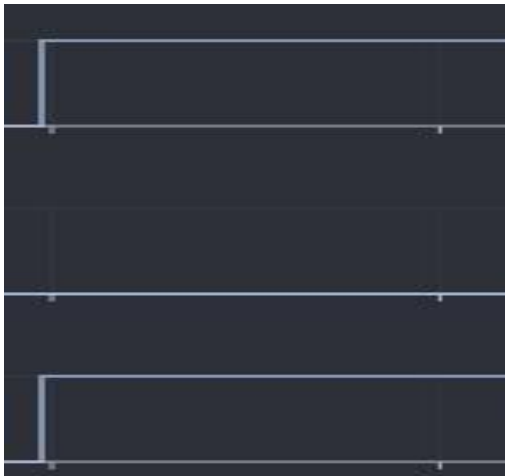
Qty	Component	Qty	Component
1	Laptop/PC		
1	OpenPLC editor software		



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III. DATA AND RESULTS

Table 1. Familiarization to OpenPLC

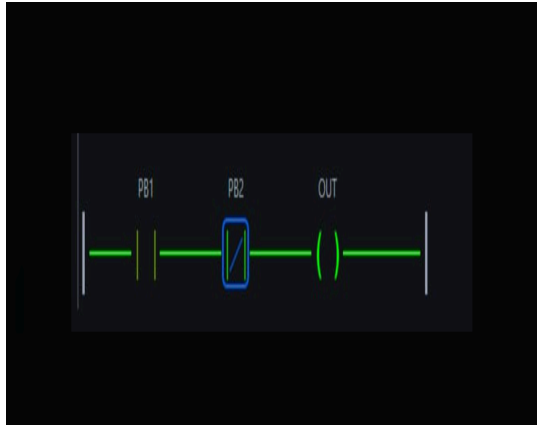
None of the Push Button is pressed:	
Ladder Diagram: 	Waveform: 
Push Button 1 is pressed:	
Ladder Diagram: 	Waveform: 



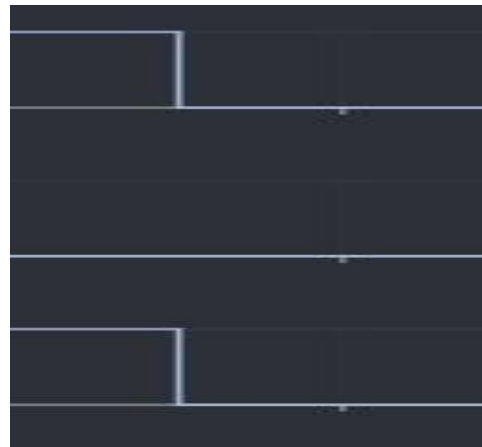
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Push Button 1 is released:

Ladder Diagram:

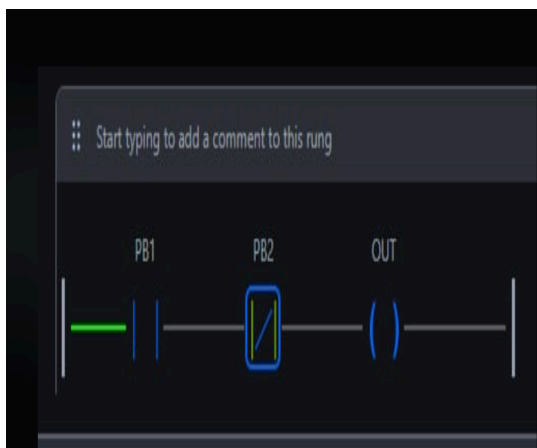


Waveform:

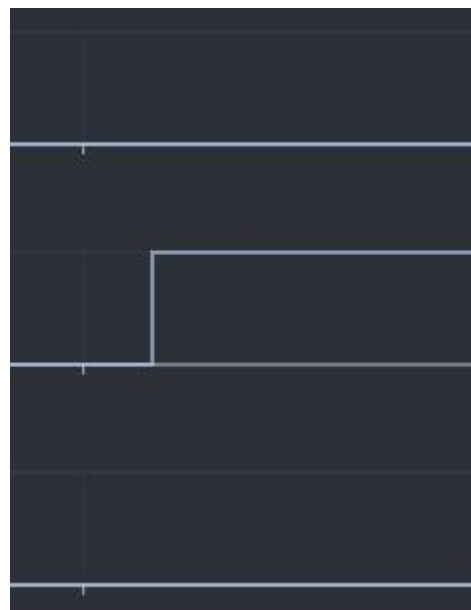


Push Button 2 is pressed:

Ladder Diagram:



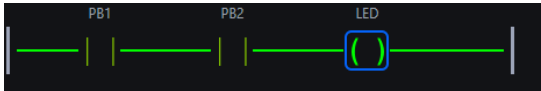
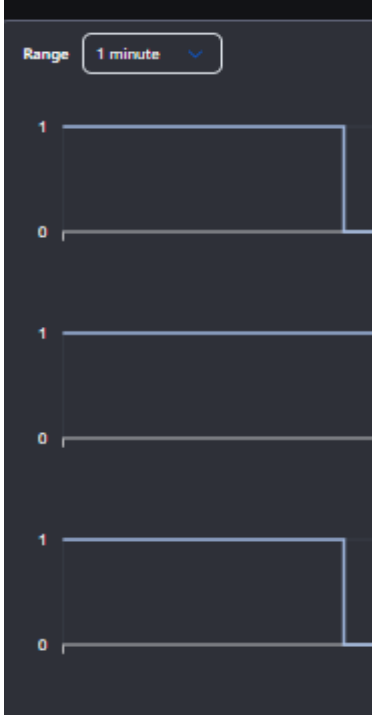
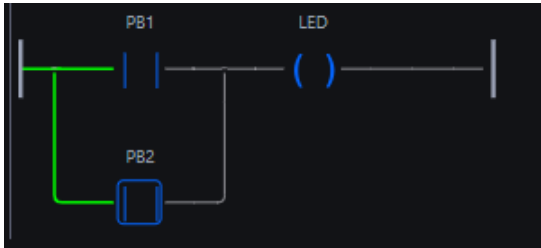
Waveform:





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Table 2. Logic Gates

AND GATE	
Ladder Diagram: 	Waveform (You only need to show one scenario no need to put all four): 
OR GATE	
Ladder Diagram: 	Waveform (You only need to show one scenario no need to put all four):

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NAND GATE	
Ladder Diagram:	Waveform (You only need to show one scenario no need to put all four):

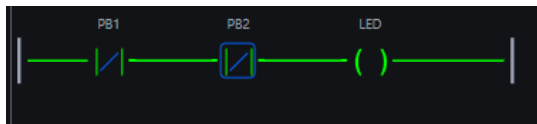


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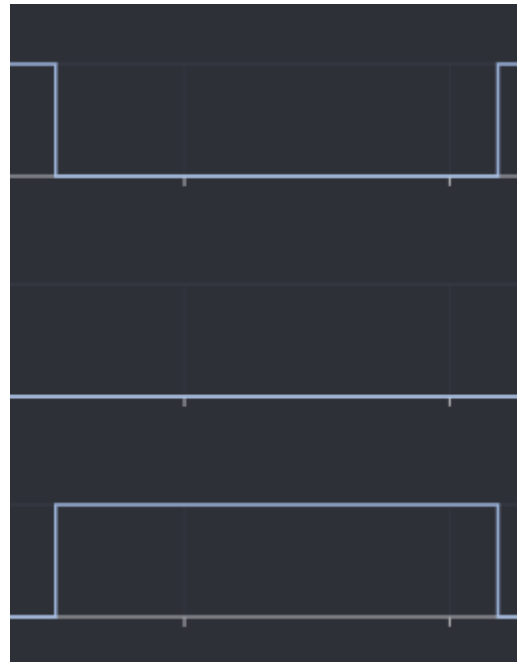
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NOR GATE

Ladder Diagram:



Waveform (You only need to show one scenario no need to put all four):

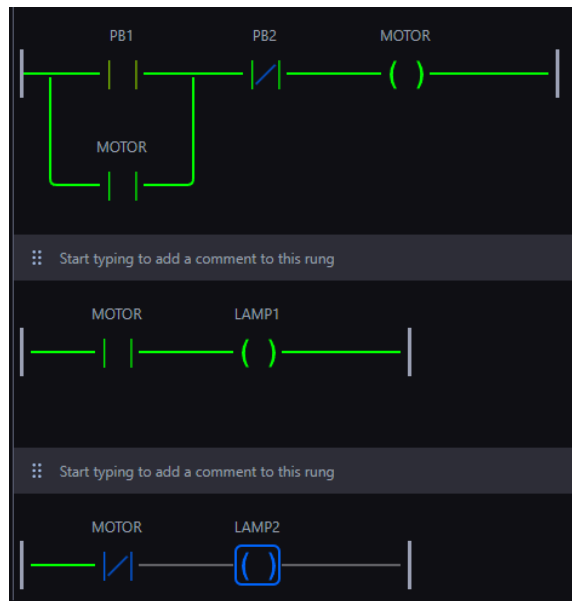




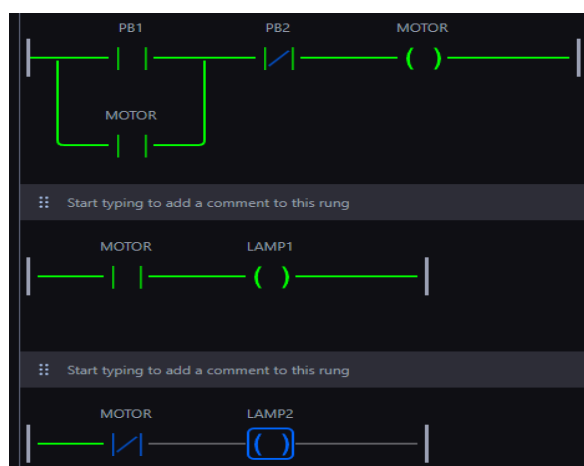
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Table 3. Motor and Lamp

PB1 is pressed:



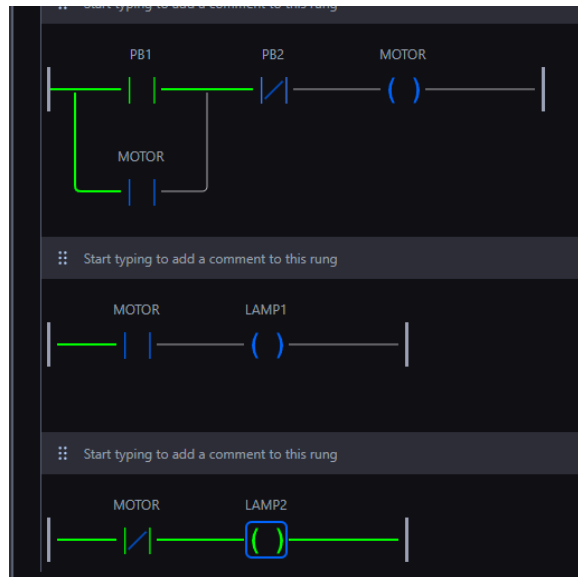
PB1 is released:





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PB2 is pressed:



IV. ANALYSIS

In the first part of the experiment, we observe the behavior of basic contacts in OpenPLC. At the start, PB1 is normally open, so no power flows and the waveforms remain flat at 0. When PB1 is pressed, the circuit closes through the normally closed PB2, energizing the OUT coil and causing the waveforms to rise to 1. Releasing PB1 opens the circuit again, bringing the output back down to 0. Since PB2 is normally closed, pressing it opens the circuit instead, cutting off power completely and keeping OUT at 0.

In the second part of the experiment, fundamental logic gates are constructed in OpenPLC using different configurations of normally open and normally closed contacts. The AND gate is formed by placing two normally open contacts in series, requiring both inputs to be actuated at the same time to complete the circuit. Meanwhile, the OR gate places the



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contacts in parallel, allowing power to reach the output if at least one input is active. To achieve inverse logic, negated or normally closed contacts are used. The NAND gate uses a parallel arrangement of negated contacts, keeping the LED energized unless both paths are broken when both buttons are pressed. On the other hand, the NOR gate wires these negated contacts in series, creating a default state of 1 that immediately drops to 0 if either button is pressed and breaks the continuity. Across all setups, the resulting waveforms confirm that the ladder routing directly follows the expected digital truth tables.

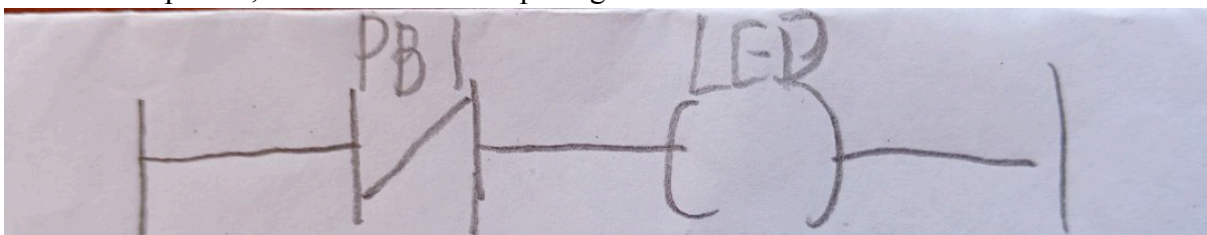
V. CONCLUSION

Write your conclusion:

VI. QUESTIONS AND PROBLEMS

1. How would you implement a NOT Gate and a XOR Gate, draw the ladder diagram.

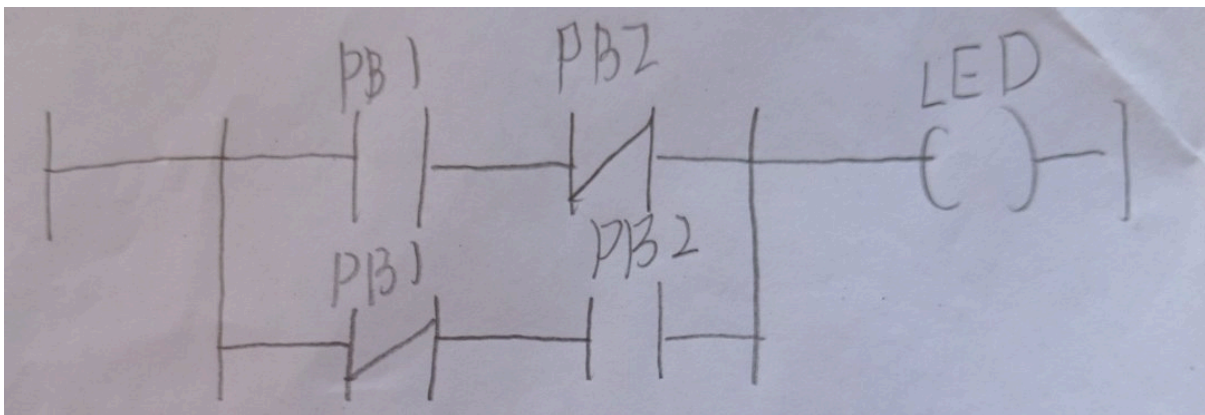
To set up a NOT gate in OpenPLC, you connect a single Normally Closed (NC) contact in series with your output coil. Since the contact is normally closed, power flows continuously, keeping the output on when the input button is not pressed. When the button is pressed, the contact opens, breaking the circuit and turning the output off, which inverts the input signal.





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To create an XOR (Exclusive OR) gate, where the output is on only when the inputs are in different states, you need a circuit with two parallel branches linked to one output coil. In the first branch, place a Normally Open (NO) contact for the first input in series with a Normally Closed (NC) contact for the second input. In the second parallel branch, reverse this setup by placing a Normally Closed (NC) contact for the first input in series with a Normally Open (NO) contact for the second input. This arrangement ensures that power reaches the output coil only if the first button is pressed while the second is released, or if the second button is pressed while the first is released.



2. Implement a stoplight system (must vary from other groups) using OpenPLC. Show the ladder diagram and link here the demonstration upon running the schematic.
(https://drive.google.com/drive/folders/1DCrXPz4udJ84QGv2xWVJTol1kfQ8KiLE?usp=drive_link)





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In this design, the contacts corresponding to the lamps act as latches and operate similarly to the motor. The negated contacts also ensure that no more than one lamp is turned on at a time. The operation works such that when a button is pressed and then released, the corresponding lamp remains open until another button is pressed and released. This requires forcing the contact to true and then forcing it to false, effectively releasing the initial contact corresponding to the lamp.